

Extensive vs. Intensive Accumulation: Choice of Technique and Capacity Utilization

Preliminary CPR Estimation Results for the US Corporate Sector

Dissertation Chapter 2

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Bottom Line Up Front (BLUF)

- **Core Question:** How do distributive struggles and capital structure condition long-run productive capacity?
- **Methodology:** We resolve the $I(2)$ cointegration trap by treating the levels relationship as a **Cointegrating Polynomial Regression (CPR)** estimated via FM-OLS and IM-OLS.
- **Multicollinearity Solution:** Residual centering (orthogonalization) stabilizes the design matrix, reducing interaction VIFs from > 80.0 to exactly **1.0**.
- **Key Empirical Result:**
 - ▶ We find a **negative and highly significant** interaction between intensive accumulation (τ) and the wage share (ω) in the Fordist era, validating the induced innovation theory on a concave frontier.
 - ▶ A stark **structural break** occurs post-1974: the interaction becomes positive and statistically non-significant, reflecting the rise of financialization and global supply chains.

The Extensive Margin (K^{NRC})

Ontological Definition

Nonresidential structures and infrastructure (K^{NRC}) represent the physical plant capacity, logistical networks, and spatial layout of production.

- **Temporal Properties:** High lumpiness, long gestation periods, and high irreversibility.
- **Asset Role:** Under the locked theory, structures act as the direct physical scale parameter for capacity ($K_t^{scale} \equiv K_t^{NRC}$).
- **Hypothesized Sign:** $\theta > 0$. Expanding physical plants directly increases the maximum potential capacity of the system.

The Intensive Margin ($\tau \equiv K^{ME}/K^{NRC}$)

Ontological Definition

The capital composition ratio (τ) represents the choice of technique—the machinery and equipment (K^{ME}) installed per unit of physical structure (K^{NRC}).

- **Temporal Properties:** Highly modular, shorter service lives, faster depreciation, and faster adaptation to technical change.
- **Asset Role:** Direct productive capacity capital that determines the technological intensity of the plant.
- **Hypothesized Sign:** $\psi > 0$. Higher machinery density increases the capacity payoff per unit of physical structure.

The Cost-Minimization FOC under Unbalanced Growth

The Transformation Frontier

Productive capacity Y^P is generated on a concave transformation frontier:

$$Y_t^P = F(K_t^{NRC}, K_t^{ME}; \omega_t) = A_t (K_t^{NRC})^\theta (K_t^{ME})^{\psi + \lambda \omega_t}$$

- **Induced Innovation:** Wage pressure (ω) forces capitalists to increase machinery density ($\tau \equiv K^{ME}/K^{NRC}$) to save on labor costs.
- **Unbalanced Gestation:** Machinery and structures cannot be substituted frictionlessly. The cost-minimization FOC yields:

$$\theta_\tau \equiv \frac{\partial \ln Y_t^P}{\partial \ln \tau_t} = \psi + \lambda \omega_t$$

- **Frontier Concavity:** Due to diminishing returns to mechanization on a concave frontier, the composition payoff must fall as the wage share rises:

$$\lambda \equiv \frac{\partial^2 \ln Y_t^P}{\partial \ln \tau_t \partial \omega_t} < 0$$

The Estimator Design Matrix

Specification A: Single-Capital Aggregate (Shaikh-style)

$$\ln Y_t = \alpha + \beta_k \ln K_t^{cap} + \beta_\omega \omega_t + u_t$$

where $K_t^{cap} \equiv K_t^{ME} + K_t^{NRC}$ is the single capital aggregate.

Specification B: Composition-Mediated (A03 Growth-Law Basis)

$$\ln Y_t = \alpha + \theta \tilde{k}_t^{scale} + \psi \tilde{\tau}_t + \phi \tilde{\omega}_t + \lambda (\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth} + \epsilon_t$$

where variables are centered around their sample means.

- **Residual Centering (Orthogonalization):** To resolve severe multicollinearity, the interaction term is replaced by the residuals of its regression on the linear terms:

$$(\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth} = (\tilde{\tau}_t \cdot \tilde{\omega}_t) - \hat{\delta}_1 \tilde{k}_t^{scale} - \hat{\delta}_2 \tilde{\tau}_t - \hat{\delta}_3 \tilde{\omega}_t$$

Cointegrating Polynomial Regression (CPR)

The $I(2)$ Cointegration Trap

Because $\tilde{\tau}_t$ and $\tilde{\omega}_t$ are both $I(1)$ processes, their product $\tilde{\tau}_t\tilde{\omega}_t$ contains $I(2)$ stochastic trends. Standard linear cointegration (e.g. Johansen) breaks down.

- **Asymptotic Corrections (FM-OLS & IM-OLS):**
 - ▶ **FM-OLS (Fully Modified OLS):** Adjusts for endogeneity and serial correlation in the residuals using a non-parametric long-run covariance matrix estimator.
 - ▶ **IM-OLS (Integrated Modified OLS):** Applies a parametric lead-and-lag correction to eliminate the endogenous feedback loop from the integrated regressors.
- **Trend Cancellation:** Cointegration is achieved because the $I(2)$ components in the interaction term cancel out with the non-linear dynamics of output, yielding an $I(0)$ stationary residual.
- **VIF Stabilization:** Residual centering drops the maximum VIF from over 80.0 in the raw interaction model to exactly **1.0**, stabilizing all parameters.

Results: Specification A (Single-Capital Baseline)

Table: Specification A: Single-Capital Aggregate (FM-OLS, Raw)

Window	Regressor	Coeff	t-stat	p-value
Golden Age (1945–1973)	$k_{Kcap,centered}$	0.9984	18.2751	0.0000
	$\omega_{NFC,centered}$	-0.5824	-0.4052	0.6881
Post-War Fordist (1940–1973)	$k_{Kcap,centered}$	0.9821	24.1122	0.0000
	$\omega_{NFC,centered}$	-0.7104	-0.5211	0.6045
Post-Fordist (1974–2024)	$k_{Kcap,centered}$	1.0543	35.8821	0.0000
	$\omega_{NFC,centered}$	-4.8824	-2.8912	0.0051

- Scale coefficient is positive and robustly close to 1.0.
- **Robustness:** Cointegration confirmed; residuals reject the null of no cointegration under both Engle-Granger and Phillips-Ouliaris tests.

Results: Specification B (Composition-Mediated)

Table: Specification B: Composition-Mediated (FM-OLS, Orthogonalized)

Window	Regressor	Coeff	t-stat	p-value
Golden Age (1945–1973)	$k_{NRC,centered}$	0.2378	3.7248	0.0010
	$\tau_{centered}$	0.3121	10.5266	0.0000
	$\omega_{NFC,centered}$	-5.1605	-3.6241	0.0014
	Interaction (orth)	-1.0294	-2.4892	0.0201
Post-War Fordist (1940–1973)	$k_{NRC,centered}$	0.0773	1.1122	0.2752
	$\tau_{centered}$	0.3809	11.7582	0.0000
	$\omega_{NFC,centered}$	-7.6639	-4.8891	0.0000
	Interaction (orth)	-2.0342	-4.9555	0.0000
Post-Fordist (1974–2024)	$k_{NRC,centered}$	0.2146	4.2083	0.0001
	$\tau_{centered}$	0.3520	16.9510	0.0000
	$\omega_{NFC,centered}$	1.1260	0.7747	0.4425
	Interaction (orth)	0.0821	0.1257	0.9006

- Composition effect (τ) is highly significant. Interaction is negative and significant during the Fordist era.
- **Robustness:** Cointegration validated by both Engle-Granger and Phillips-Ouliaris tests, ensuring stationary residuals.

Critical Discussion: Validation vs. Structural Crisis

Validation of the Theory

The negative and highly significant interaction coefficient ($\lambda < 0$) in the Fordist era (1940 – 1973) strongly validates the induced innovation theory. Higher wage shares (ω) force capitalists to mechanize, but this drives the economy into the concave zone of lower marginal capacity conversion.

The Post-Fordist Structural Break

Post-1974, the interaction collapses to zero (0.0821, p-value 0.90).

- **Global Offshoring:** Domestic wage bargaining no longer drives technique choice in the same way because capital can relocate.
- **Financialization:** Capital accumulation shifts away from real plant investment to financial channels.
- **Off-shoring/Subcontracting:** The domestic wage share ω_t ceases to act as a reliable proxy for the total cost of labor involved in global production chains.

Unpacking Specification B: Golden Age Case Study (1945–1973)

Estimated Cointegrating Equation (FM-OLS)

$$y_t = 14.8617 + 0.2378\tilde{k}_t^{scale} + 0.3121\tilde{\tau}_t - 5.1605\tilde{\omega}_t - 1.0294(\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth}$$

- **State-Dependent Capacity Payoff of Composition (θ_t^τ):**

$$\theta_t^\tau \equiv \frac{\partial y_t}{\partial \tilde{\tau}_t} = 0.3121 - 1.0294\tilde{\omega}_t$$

- **Numerical Sensitivities of Wage Share Variations:**

- ▶ **Baseline (Average ω_t):** $\tilde{\omega}_t = 0 \implies \theta_t^\tau = \mathbf{0.3121}$.
- ▶ **High Wage Share ($\tilde{\omega}_t = +0.02$):** $\theta_t^\tau = 0.3121 - 1.0294(0.02) = \mathbf{0.2915}$
(payoff falls by $\approx 6.6\%$).
- ▶ **Low Wage Share ($\tilde{\omega}_t = -0.02$):** $\theta_t^\tau = 0.3121 - 1.0294(-0.02) = \mathbf{0.3327}$
(payoff rises by $\approx 6.6\%$).

- **Economic Intuition:** Higher distributive wage pressure forces capitalists to accelerate mechanization, but drives the system into a concave region of the frontier, lowering the marginal capacity payoff of machinery.

Capacity Utilization & Elasticity Reconstruction (1945–1973)

- **Sectoral Division:** Long-run elasticities estimated on productive-origin NFC data, but utilization (μ_t) is reconstructed relative to **Total Real GDP** (observed aggregate demand).
- **Historical Path Comparison:**
 - ▶ **Specification B (Composition-Mediated):** Stable operating corridor ($\sigma_\mu = 12.98\%$). Anchored at 1973 peak ($\mu_{1973} = 1.0$).
 - ▶ **Shaikh-Style:** Artificially volatile cycle ($\sigma_\mu = 15.06\%$) due to the constant scale assumption.
 - ▶ **HP Filter:** Severely dampens real capacity squeezes ($\sigma_\mu = 3.79\%$) by treating demand booms (e.g., WWII/Korean War) as shifts in trend capacity.
- **Methodological Agreement:** Spec B utilization has high positive correlation with Shaikh-style ($\rho = 0.6974$) and the HP filter cycle ($\rho = 0.4024$).
- **Mechanization Bias & Cushioning:**
 - ▶ Structures elasticity is negative ($\theta_t^{NRC} \approx -0.07$), indicating that building empty plants without machinery has a negative direct capacity payoff.
 - ▶ Growth-weighted aggregate transformation elasticity ($\theta_t^{total} = g_{Y\rho}/g_K^N$) is positive and dynamic (**0.84 to 2.30**).
 - ▶ In the 1960 slowdown, Spec B remains high ($\mu_{1960} = 1.19$), cushioning the collapse seen in Shaikh ($\mu_{1960} = 0.86$) through slow capacity envelope

Conclusion & Next Steps

- We have established a robust, cointegrated long-run relationship using Cointegrating Polynomial Regression (CPR) and residual centering.
- The results confirm that capital composition (τ) mediates capacity conversion, and that this mediation is conditioned by distribution (ω) in the Fordist core.
- **Reconstructing Capacity & Utilization:**
 - ▶ Reconstruct the potential capacity output series:

$$\hat{y}_t^p = \hat{\alpha} + \hat{\theta} \tilde{k}_t^{scale} + \hat{\psi} \tilde{\tau}_t + \hat{\phi} \tilde{\omega}_t + \hat{\lambda} (\tilde{\tau}_t \tilde{\omega}_t)_{orth}$$

- ▶ Recover the latent capacity utilization series as the stationary cointegrating residual:

$$\widehat{\ln \mu_t} = y_t - \hat{y}_t^p$$